MATHEMATICS

REVISION PLAN-3

Target : JEE (Advanced)-2023

DPP
DAILY PRACTICE PROBLEM
DPP NO. # 03 (Advanced)

Max. Time : 129 Minutes			Total Marks-166
Q. 01 to 06	Comprehension	('–1' negative marking)	[3 Marks,3 Min.][18,18]
Q. 07 to 26	Multiple choice objective	('–1' negative marking)	[4 Marks,3 Min.][80,60]
Q. 27 to 28	Match the column	(no negative marking)	[4 Marks,3 min.][8,6]
Q. 29 to 43	Numerical Type	('–1' negative marking)	[4 Marks,3 Min.][60,45]

DPP Syllabus : Application of Derivatives, Functions & ITF and Limits, Continuity & Derivability

Comprehension # 1 (Q.No. 1 to 2)

Let $f : R \to R$, $y = f(x)$, $f(0) = 0$, $f'(x) > 0$ and $f''(x) > 0$. Three points $A(\alpha, f(\alpha))$, $B(\beta, f(\beta))$, $C(\gamma, f(\gamma))$ on $y = f(x)$ such that $0 < \alpha < \beta < \gamma$.

1. Which of the following is false?

(A) $\alpha f(\beta) > \beta(f(\alpha))$ (B) $\alpha f(\beta) < \beta(f(\alpha))$ (C) $\gamma f(\beta) < \beta(f(\gamma))$ (D) $\gamma f(\alpha) < \alpha(f(\gamma))$

2. Which of the following is true?

(A) $\dfrac{f(\alpha)+f(\beta)}{2} < f\left(\dfrac{\alpha+\beta}{2}\right)$ (B) $\dfrac{f(\alpha)+f(\beta)}{2} > f\left(\dfrac{\alpha+\beta}{2}\right)$

(C) $\dfrac{f(\alpha)+f(\beta)}{2} = f\left(\dfrac{\alpha+\beta}{2}\right)$ (D) $\dfrac{2f(\alpha)+f(\beta)}{3} < f\left(\dfrac{2\alpha+\beta}{3}\right)$

Comprehension # 2 (Q.No. 3 to 4)

Let $f : (0, \infty) \to (0, \infty)$ be differentiable function such that $f'\left(\dfrac{a}{x}\right) = \dfrac{x}{f(x)}$, where a is positive constant and $f'(1) = 1$ and $f'(2) = 2$. Then

3. The value of $\dfrac{f(2)}{f(1)}$ is equal to

(A) $\dfrac{1}{2a}$ (B) $\dfrac{1}{a}$ (C) a (D) 2a

4. $\lim\limits_{n\to\infty}\left\{f(n)\displaystyle\int_0^{\frac{1}{n}} x^{x+1}dx\right\}$ is equal to

(A) $\dfrac{1}{2a}$ (B) $\dfrac{1}{a}$ (C) a (D) 2a

Comprehension # 3 (Q.No. 5 to 6)

Let f(x) be a function such that its derivative $f'(x)$ is continuous in [a, b] and differentiable in (a, b). Consider a function $\phi(x) = f(b) - f(x) - (b - x) f'(x) - (b - x)^2 A$. If Rolle's theorem is applicable to $\phi(x)$ on [a. b], answer following questions.

5. If there exists some number $c(a < c < b)$ such that $\phi'(c) = 0$ and $f(b) = f(a) + (b - a) f'(a) + \lambda(b - a)^2 f''(c)$, then λ is

(A) 1 (B) 0 (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

6. Let $f(x) = x^3 - 3x + 3$, a = 1 and b = 1 + h. If there exists $c \in (1, 1 + h)$ such that $\phi'(c) = 0$ and $\frac{f(1+h)-f(1)}{h^2} = \lambda c$, then λ =

(A) $\frac{1}{2}$ (B) 2 (C) 3 (D) does not exist

7. If parametric equation of any curve at θ is given by $x = a\cos^3\theta$, $y = a\sin^3\theta$, then $a > 0$, $\theta \in \left(0, \frac{\pi}{2}\right)$

(A) length of tangent is $a\sin^2\theta$
(B) length of subtangent is $a\sin^2\theta\cos\theta$
(C) length of normal is $a\tan\theta\sin^2\theta$
(D) length of subnormal is $a\sin^3\theta\cot\theta$

8. Which of the following pair(s) of family is/are orthogonal?

(A) $16x^2 + y^2 = c$ and $y^{16} = kx$
(B) $y = x + ce^{-x}$ and $x + 2 = y + ke^{-y}$
(C) $y = cx^2$ and $x^2 + 2y^2 = k$
(D) $x^2 - y^2 = c$ and $xy = k$

where c and k are arbitrary constant

9. If $f''(x) > 0$, $\forall x \in R$, $f'(3) = 0$ and $g(x) = f(\tan^2 x - 2\tan x + 4)$, $0 < x < \frac{\pi}{2}$, then g(x) is increasing in

(A) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$ (B) $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ (C) $\left(\frac{\pi}{4}, \frac{\pi}{3}\right)$ (D) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

10. Let $f(x) = (x + 1)\ \ell n(1+x)$, $g(x) = x + \frac{x^2}{2}$ and $h(x) = x + \frac{x^2}{2} - \frac{x^3}{3}$, then for $x > 0$.

(A) $f(x) < g(x)$ (B) $f(x) > g(x)$ (C) $f(x) < h(x)$ (D) $f(x) > h(x)$

11. Suppose f is twice differentiable function satisfying $f(0) = 2$, $f'(0) = 11$, $f''(0) = -8$, $f(2) = 5$, $f'(2) = -3$, $f''(2) = 7$ and also suppose that the function $g(x) = \frac{1}{x}\int_0^x f(t)\,dt$ has a critical point at x = 2. Then

(A) Critical point of g(x) at x = 2 is local minima
(B) Critical point of g(x) at x = 2 is local maxima
(C) $\int_0^2 f(t)dt = 10$
(D) $\int_0^2 f(t)dt = 20$

Reliable INSTITUTE
Address : 'Reliable Institute', A-10 Road No.1, IPIA, Kota-324005 (Rajasthan), INDIA
visit us at: www.reliablekota.com, Email: info@reliablekota.com
Call us: +91-744-2665544

12. Let $f:\left[-\frac{5\pi}{12},-\frac{\pi}{3}\right]\to R$ is a function $f(x)=\tan\left(x+\frac{2\pi}{3}\right)-\tan\left(x+\frac{\pi}{6}\right)+\cos\left(x+\frac{\pi}{6}\right)$ then :

(A) Maximum value of f(x) is $\frac{11}{6}\sqrt{3}$

(B) Minimum value of f(x) is $\frac{2\sqrt{2}+1}{\sqrt{2}}$

(C) f(x) is a monotonic decreasing function in it's domain

(D) f(x) is a monotonic increasing function in it's domain.

13. If P is any point on the curve $x^2 + 3y^2 + 3xy = 1$ whose centre is at O, then is

(A) maximum value of OP is $\sqrt{\frac{2}{\sqrt{13}-2}}$ (B) minimum value of OP is $\sqrt{\frac{2}{\sqrt{13}+2}}$

(C) minimum value of OP is $\sqrt{\frac{2}{4+\sqrt{13}}}$ (D) maximum value of OP is $\sqrt{\frac{2}{4-\sqrt{13}}}$

14. Let $f : R \to R$, $f(x) = x + \log_e(1 + x^2)$. Then

(A) f is injective

(B) f is surjective

(C) there is a point on the graph of y = f(x) whose tangent is not parallel to any of the chords

(D) Inverse of f(x) exists

15. Let f, g and h be three functions defined as follows:

$$f(x) = \frac{32}{4+x^2+x^4},\ g(x) = 9 + x^2 \text{ and } h(x) = -x^2 - 3x + k.$$

Identify which of the following statement(s) is (are) correct?

(A) Number of integers in the range of f(x) is 8.

(B) Number of integral values of k for which $h(f(x)) > 0$ and $h(g(x)) < 0\ \forall\ x \in R$ is 20.

(C) Number of integral values of k for which $h(f(x)) > 0$ and $h(g(x)) < 0\ \forall\ x \in R$ is 19.

(D) Maximum value of g(f(x)) is 73.

16. Let f(x, y) be a periodic function satisfying the condition $f(x, y) = f(2x + 2y, 2y - 2x)\ \forall\ x, y \in R$ and let g(x) be a function defined as $g(x) = f(2^x, 0)$. Then period of g(x) is/are

(A) 6 (B) 12 (C) 24 (D) 48

17. Let $f : R \to [-1, 1]$ and $g : R \to B$, where R be the set of all real numbers and $g(x) = \sin^{-1}\left(\frac{f(x)}{2}\sqrt{4-f^2(x)}\right)+\frac{\pi}{3}$. If y = f(x) and y = g(x) are both surjective, then set B lies in

(A) $\left[-\frac{\pi}{3},\frac{\pi}{3}\right]$ (B) $\left[0,\frac{2\pi}{3}\right]$ (C) $\left[0,\frac{\pi}{3}\right]$ (D) $[0, \pi]$

18. Let a = $\sin^{-1}(\sin 3) + \sin^{-1}(\sin 4) + \sin^{-1}(\sin 5)$, $f(x) = e^{x^2+|x|}$, domain of f(x) be $[a, \infty)$, range of f(x) be $[b, \infty)$, $g(x) = (4\cos^4 x - 2\cos 2x - \frac{1}{2}\cos 4x - x^7)^{1/7}$, domain and range of g(x) is set of real numbers. Which of the following are CORRECT ?

(A) $a = -2$
(B) $b + a = -1$
(C) $f(gog(b)) = e^2$
(D) both f(x), g(x) are non-invertible functions

19. If the least and the greatest values of 'x' satisfying the inequation $[\sin^{-1}x] > [\cos^{-1}x]$, (where [.] denotes the greatest integer function) are p and q respectively, then

(A) the point (2, cos1 + 1) lies on the line y = p + q
(B) $p + q > 1 + \sin 5$
(C) $\log_{q-p} \pi$ is negative
(D) $\cos^{-1} p < \tan^{-1} q$

20. The value of $\tan^{-1}\frac{4}{7} + \tan^{-1}\frac{4}{19} + \tan^{-1}\frac{4}{39} + \tan^{-1}\frac{4}{67} + \ldots\ldots \infty$ is

(A) $\tan^{-1}1 + \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}$
(B) $\tan^{-1}1 + \cot^{-1}3$
(C) $\cot^{-1}1 + \cot^{-1}\frac{1}{2} + \cot^{-1}\frac{1}{3}$
(D) $\tan^{-1}2$

21. If $k \in N$ and $a \in (0, \infty) - \{1\}$ then $\lim_{n\to\infty} n^k (a^{\frac{1}{n}} - 1)\left(\sqrt{\frac{n-1}{n}} - \sqrt{\frac{n+1}{n+2}}\right)$

(A) 0 for exactly 3 values of k
(B) Non zero finite quantity for exactly 2 values of k
(C) 0 for exactly 2 values of k
(D) Non zero finite quantity for exactly 1 values of k

22. If x = a satisfies $\tan^{-1}(x+2) + \cot^{-1}\sqrt{4x+20} = \lim_{x\to\frac{\pi}{2}} \frac{\sin(x\cos x)}{\cos(x\sin x)}$ and $\lim_{x\to 0}\left(\frac{(1+x)^{\frac{1}{a}} - \frac{(1+x)^{\frac{1}{x}}}{e} - bx}{5x + kx^2 + x^3}\right) = 0$ then

(A) a = 2
(B) ab = 3
(C) a = 1 ,b = 3
(D) $k \in R$

23. If $\ell(x) = \frac{e^x + e^{-x}}{2 + x^2}$, $f(x) = \lim_{r\to\infty}\left(\left(\ell(x)^r\right) + \beta^r\right)^{\frac{1}{r}}$ where $0 < \beta < 1$ (constant), then

(A) f(x) is constant function
(B) f(1) = f(–1)
(C) f(1) + f(–1) = 0
(D) f(x) is non constant function

24. Let $f : R \to R$ be a function, such that $|f(x)| \le x^{4n}$, $n \in N\ \forall\ x \in R$ then f(x) is:

(A) discontinuous at $x = 0$ (B) continuous at $x = 0$

(C) non differentiable at $x = 0$ (D) differentiable at $x = 0$

25. Let $f(x) = x^2 - 2|x|$ and $g(x) = \begin{cases} \min\{f(t) & -2 \le t \le x \quad -2 \le x < 0 \\ \max\{f(t) & 0 \le t \le x \quad 0 \le x \le 2 \\ f(x) & x > 2 \end{cases}$, then g(x) is not differentiable at

(A) $x = 0$ (B) $x = -1$ (C) $x = 2$ (D) $x = 1$

26. Let f be continuous on [0,1] and differentiable on (0,1). Suppose that $f(0) = f(1) = 0$ and that there is $c \in (0,1)$ such that $f(c) = 1$, then

(A) $|f'(d)| > 3$ for some $d \in (0, 1)$ (B) $|f'(d)| > 2$ for some $d \in (0, 1)$

(C) $|f'(d)| > \frac{3}{2}$ for some $d \in (0, 1)$ (D) $|f'(d)| < 3\ \forall\ d \in (0, 1)$

27. A function f(x) is defined as $f(x) = \begin{cases} 6x - 5 - x^2 & ; \quad x \le 3 \\ 24x - 32 - 4x^2 & ; \quad x > 3 \end{cases}$.

Tangents are made on f(x) in the first quadrant. Let tangent $T_1 : y = m_1x + b_1$ and $T_2 : y = m_2x + b_2$ have highest and lowest y-intercepts of all tangents made in first quadrant. Then match the following

(A)	m_1	(p)	32
(B)	m_2	(q)	–4
(C)	b_1	(r)	–8
(D)	b_2	(s)	4

28. Let $x, y, z \in R$ satisfying the system, of equations $x + [y] + \{z\} = 12.7$, $[x] + \{y\} + z = 4.1$ and $\{x\} + y + [z] = 2$ (where {.} and [.] denotes the fractional and integral parts respectively). Then match the following

(A)	$\{x\} + \{y\} =$	(p)	7.7
(B)	$[z] + [x] =$	(q)	1.1
(C)	$x + \{z\} =$	(r)	1
(D)	$z + [y] - \{x\} =$	(s)	3
		(t)	4

29. Let A be the point where the curve $5\alpha^2x^3 + 10\alpha x^2 + x + 2y - 4 = 0$ ($\alpha \in R, \alpha \ne 0$) meets y-axis. Then the equation of tangent to the curve at the point where the normal at A meets the curve again, is $ax + by + 2 = 0$. The value of $a - b$ is

30. If $y = (x^3 - c)$ and $y = x^2 + ax + b$ touch each other at the point (1,2) then find the value of $\frac{|a| + |b| + |c|}{|a + b - c|}$.

31. Let f be be a continuous and differentiable function in (x_1, x_2). If $f(x) \,.\, f'(x) \geq x\sqrt{1-(f(x))^4}$ and $\lim\limits_{x\to x_1^+}(f(x))^2 = 1$ and $\lim\limits_{x\to x_2^-}(f(x))^2 = \frac{1}{2}$, then minimum value of $\left(\frac{x_1^2 - x_2^2}{\pi}\right)$ is

32. Let $\theta(x)$ denotes an angle measured in radians, subtended by a fixed closed interval [1,3] on the y-axis at a point (x,0) on the positive x-axis . If θ_0 is the maximum value of θ (x) then find the value $\frac{5\pi}{4\theta_0}$

33. From a fixed point A on the circumference of a circle of radius 2, the perpendicular AY is let fall on the tangent at any point P. The maximum area of the triangle APY is. (Take $\sqrt{3}$ = 1.73)

34. Let A = {1, 2, 3, 4} and B = {3, 4, 5, 6, 7} are two sets. Let m is the number of one-one functions $f : A \to B$ such that $f(i) \neq i \ \forall \ i \in A$ and n is the number of one-one function $f : A \to B$ such that $|f(i) - i| \leq 3 \ \forall \ i \in A$, then find the value of (m + n).

35. Let $f : N \to [0, \infty)$ be a function satisfying the following conditions :

(a) f(100) = 10

(b) $\frac{1}{f(0)+f(1)} + \frac{1}{f(1)+f(2)} + \frac{1}{f(2)+f(3)} + \ldots\ldots.. + \frac{1}{f(n)+f(n+1)} = f(n+1)$, for all non-negative integers n.

Find the value of f(9801)

36. Let $\left(1, \frac{1}{4}\right)$ be vertex of quadratic polynomial y = f(x) passing through (0,0) and $y = f^{-1}(x)$ be inverse of y = f(x), $x \in (-\infty, 1]$. If A is the area bounded by y = f(x), $y = f^{-1}(x)$ and line 4x + 4y – 5 = 0 then 96A is equal to

37. For $y = \cot^{-1}\left(\frac{\log ex^2}{\log e/x^2}\right) + \tan^{-1}\left(\frac{3+2\ln x}{1-6\ln x}\right)$, the value of $\frac{d^2y}{dx^2}$ at x = 1, is equal to ______

38. If $2^{2\pi/\sin^{-1}x} - 2(a+2)\,2^{\pi/\sin^{-1}x} + 8a < 0$ for atleast one real x and non negative value of $a \in [\alpha, \beta) \cup (\gamma, \infty)$, then find $\alpha + \beta + \gamma$

39. If $L = \lim\limits_{k\to\infty} \prod\limits_{n=3}^{k}\left(1 - \frac{4}{n^2}\right)$, $N = \lim\limits_{k\to\infty} \prod\limits_{n=1}^{k} \frac{(1+n^{-1})^2}{(1+2n^{-1})}$ then the value of $\frac{1}{L} + \frac{1}{N}$ is

40. If $\lim\limits_{x\to 0}\left(\frac{1-\sqrt{\cos 2x}\,\sqrt[3]{\cos 3x}\,\sqrt[4]{\cos 4x}\ldots\ldots\ldots.\sqrt[n]{\cos nx}}{x^2}\right)$ is equal to 10, then n =

41. If $\lim\limits_{n\to\infty}\left(\dfrac{\left(1+\dfrac{1}{n}\right)^{n}}{\left(1-\dfrac{1}{n}\right)^{n}}-e^{2}\right)n^{2}=\dfrac{ae^{2}}{b}$ (a, b $\in$ N). Find the minimum value of (a + b).

42. Let f(x) = $\begin{cases}\sqrt{(2n+1)x-x^{2}-(n^{2}+n)}, & n\le x<n+\dfrac{1}{2}\\ (n+1)-x, & n+\dfrac{1}{2}\le x<n+1\end{cases}$ (n $\in$ I)

Find the number of values of x where f(x) is non derivable in (–5, 5).

43. Let $f(x+2y)=f(x)(f(y))^{2}$ for all x and y. If $f'(0)=\ln 2$ and f(x) is differentiable

f(x) + f(2x) + f(3x) + ………+ f(nx) $=\dfrac{Af(x)\left(f\left(nx\right)+B\right)}{f(x)+C}$

Then find the value of $\dfrac{10(A^{2}+B^{2}+C^{2})}{(3A+B+C)}$ (where A,B,C $\in$ R, f(x) $\neq$ 0)

ANSWER KEY

1.	(B)	**2.**	(B)	**3.**	(D)	**4.**	(A)	**5.**	(C)	**6.**	(C)	**7.**	(ABC)
8.	(ABCD)			**9.**	(ACD)	**10.**	(AD)	**11.**	(BC)	**12.**	(ABC)	**13.**	(CD)
14.	(ABCD)			**15.**	(ACD)	**16.**	(BCD)	**17.**	(BD)	**18.**	(ABC)	**19.**	(BCD)
20.	(BD)	**21.**	(CD)	**22.**	(BD)	**23.**	(BD)	**24.**	(BD)	**25.**	(AC)	**26.**	(BC)
27.	(A) $\rightarrow$ r, (B) $\rightarrow$ s, (C) $\rightarrow$ p, (D) $\rightarrow$ q					**28.**	(A) $\rightarrow$ r, (B) $\rightarrow$ s, (C) $\rightarrow$ p, (D) $\rightarrow$ q						
29.	(3)	**30.**	(1.00)	**31.**	(0.33)	**32.**	(7.50)	**33.**	(2.59 or 2.60)			**34.**	(94)
35.	(99)	**36.**	(37)	**37.**	(0)	**38.**	(2.12 or 2.13)			**39.**	(6.5)	**40.**	(6)
41.	(5)	**42.**	(19)	**43.**	(30.00)								

Reliable INSTITUTE
Address : 'Reliable Institute', A-10 Road No.1, IPIA, Kota-324005 (Rajasthan), INDIA
visit us at: www.reliablekota.com, Email: info@reliablekota.com
Call us: +91-744-2665544